

Advanced Mathematical Tools for Management

MATH 40604 — Study Notes

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How to use these notes

These notes accompany **MATH 40604, Advanced Mathematical Tools for Management**. They are a study aid, not a replacement for the lectures: the official course material is the slide deck distributed each week. What the slides compress — the intermediate algebra, the reason a definition is written the way it is, the mistake that costs marks — is what these notes expand.

They are written on one assumption: that you are a capable adult who has not done mathematics in a while. Nothing here requires you to remember a formula from years ago. Where a step looks small, it is written out anyway.

The structure of each chapter

Every chapter follows the same path.

- **Definitions** state an idea precisely, and are always followed by an *In words* reading in plain English. If the symbols do not land, read the plain-English version first and come back.
- **Key results** are the facts you will actually use. Each one is derived, not asserted.
- **Worked examples** are set in a business or analytics context and show every algebraic step.
- **Common pitfalls** collect the errors that recur — usually not from misunderstanding the concept, but from a small slip in notation.
- **One-minute recap** is what to reread on the train.

Notation used throughout

Mathematical notation varies between textbooks, countries, and software. The conventions below are fixed for the whole course; where a different convention is common elsewhere, it is noted.

Symbol	Meaning	Note
$\mathbb{N}$	natural numbers $\{0, 1, 2, \dots\}$	includes zero in this course
$\mathbb{Z}$	integers $\{\dots, -1, 0, 1, \dots\}$	
$\mathbb{Q}$	rational numbers	quotients p/q with $q \neq 0$
$\mathbb{R}$	real numbers	
$\mathbb{R}_+$	non-negative reals, $[0, +\infty)$	includes 0
$\mathbb{R}^p$	real vectors with p components	
$\mathbb{R}^{n \times p}$	real matrices, n rows and p columns	
(a, b)	open interval	parentheses , not $]a, b[$
$[a, b]$	closed interval	

Symbol	Meaning	Note
Δ_p	the simplex in $\mathbb{R}^p$	p components, non-negative, summing to 1
$\mathbf{x}$	a vector	bold , always
n	number of observations	never anything else
p	number of variables	never anything else
$\bar{x}$	mean of an observed sample	
μ	mean of a population or distribution	not the same object as $\bar{x}$
$\ln$	natural logarithm, base e	written $\ln$, never $\log$
$\log_{10}$	logarithm base ten	always written with its base

Four of these are worth pausing on, because they are the ones that silently change answers.

Zero is a natural number here. Many textbooks, particularly North American ones, start $\mathbb{N}$ at 1. This course starts at 0. So “ $0 \in \mathbb{N}$ ” is *true* in this course and false in some books you may own. When in doubt, look at the convention table rather than at your memory.

Open intervals use parentheses. French-language material writes the open interval from a to b as $]a, b[$. This course writes (a, b) , the convention used in statistics and data science software. You will see the French form on some slides; it means the same thing.

“Non-negative” and “positive” are different words. *Positive* means strictly greater than zero. *Non-negative* means greater than or equal to zero. French uses *positif* for both, so translated material sometimes says “positive” where “non-negative” is meant. When it matters, this course says which.

$\ln$ and $\log$ are not interchangeable. In R, $\log()$ is the *natural* logarithm and $\log_{10}()$ is base ten. In many spreadsheets and in engineering texts, **LOG** means base ten. These notes always write $\ln$ for the natural logarithm and always write the base explicitly otherwise.

A note on effort

You will get more out of a worked example by covering the solution and attempting it than by reading it. The examples are chosen so that this is realistic: none of them requires an insight you have not been given.

1 Modelling, Sets, and Algebra

This week at a glance

By the end of this chapter you should be able to:

- take a described business situation and say which quantities are **variables** and which are **parameters**, and justify the split;
- write down the objective and constraints of a small decision problem in symbols;
- describe a set three ways — by listing, by a defining property, and as an interval — and move between them;
- read and write $\cup$, $\cap$, complement, and difference, and compute the size of a union without double-counting;
- expand and condense $\sum$ notation, and prove the two identities about deviations from the mean;
- solve linear and quadratic equations, and use logarithms to turn a multiplicative relationship into a linear one.

Nothing here is new mathematics. Most of it you have met before and put down. The aim is to pick it up in a form you can use.

2 What a mathematical model is

A **mathematical model** is a deliberately simplified description of a real situation, written in symbols, built to answer a specific question. The word *deliberately* is doing real work: a model leaves things out on purpose. A model of a delivery network that accounted for every pothole would be useless, because you could not solve it and could not learn anything from it.

Three questions organise every model you will meet in this course:

1. What am I allowed to choose, or what am I trying to learn?
2. What am I taking as given?
3. What has to be true for the answer to be believable?

The vocabulary below is just a precise way of asking those three questions.

2.1 Variables and parameters

Almost every difficulty students have with modelling notation traces back to blurring these two words, so it is worth a sharp test rather than a description.

i Definition: variable and parameter

Fix a model and a question you are asking of it.

A **parameter** is a quantity whose value is settled *before* you solve the model. It is an input: given in the problem statement, measured, or estimated from data.

A **variable** is a quantity whose value is *produced by* solving the model, or supplied by an observation. It is not settled in advance.

In words: parameters are what you feed in; variables are what comes out, or what varies across the rows of your data.

! Key result: the operative test

To classify a quantity, ask one question:

Do I already know this number before I start solving?

If yes, it is a parameter. If no, it is a variable.

This test is more reliable than any list of examples, because the *same symbol can switch roles depending on the question*. That is not a flaw in the definitions; it is the point.

Consider a straight-line relationship $y = \beta_0 + \beta_1 x$.

- If you are a manager who has been handed $\beta_0 = 12$ and $\beta_1 = 3$ and asked to predict y when $x = 20$: then β_0, β_1 are **parameters** and y, x are **variables**.
- If you are an analyst with a spreadsheet of (x, y) pairs, trying to work out what β_0 and β_1 are: then the data (x, y) are the inputs and β_0, β_1 are what you are solving for. The roles have swapped.

Same equation, same letters, different question. Keep the question in view and the classification is never ambiguous.

2.2 The components of a decision model

When the question is “*what should we do?*”, the model has a standard anatomy. The example below carries the vocabulary; read the story first, then the table.

Worked example: staffing a courier fleet

A grocery-delivery startup plans next week’s courier capacity. It can use two kinds of shift.

- A **salaried shift** completes 40 deliveries and costs \$180.
- A **gig shift** completes 15 deliveries and costs \$80.

The week’s forecast demand is 600 deliveries, and all of it must be covered. A labour agreement caps salaried shifts at 12 per week. The gig platform can supply at most 20 shifts. The company wants to cover demand as cheaply as possible.

Setting it up. What can the company choose? The number of each kind of shift. Nothing else in the description is up to them — the costs, the capacities, the forecast, and the two caps are all handed over. So:

$$x_S = \text{number of salaried shifts}, \quad x_G = \text{number of gig shifts}.$$

Total cost is 180 per salaried shift plus 80 per gig shift:

$$\text{minimise } 180x_S + 80x_G.$$

Deliveries completed must reach 600:

$$40x_S + 15x_G \geq 600.$$

And the two supply caps, plus the obvious requirement that you cannot schedule a negative number of shifts:

$$x_S \leq 12, \quad x_G \leq 20, \quad x_S \geq 0, \quad x_G \geq 0.$$

Solving it. Compare the two shift types on cost per delivery:

$$\frac{180}{40} = \$4.50 \text{ per delivery}, \quad \frac{80}{15} \approx \$5.33 \text{ per delivery}.$$

Salaried shifts are cheaper per delivery, so use as many as the agreement permits: $x_S = 12$. Those cover

$$40 \times 12 = 480 \text{ deliveries,}$$

leaving $600 - 480 = 120$ to be covered by gig shifts. Each covers 15, so

$$x_G \geq \frac{120}{15} = 8.$$

Taking $x_G = 8$ (there is no reason to buy more than needed), the cost is

$$180(12) + 80(8) = 2160 + 640 = \$2,800.$$

Check the answer is legal. Deliveries: $40(12) + 15(8) = 480 + 120 = 600 \geq 600$, satisfied. Salaried shifts: $12 \leq 12$, satisfied. Gig shifts: $8 \leq 20$, satisfied. Both variables non-negative. All four constraints hold, so the schedule is feasible.

Now the anatomy, using that example:

Component	What it is	In the courier example
Decision variables	The quantities you choose. Unknown before solving.	x_S, x_G
Parameters	Values handed to you. Known before solving.	costs 180 and 80; capacities 40 and 15; demand 600; caps 12 and 20
Objective function	The single number you are making as large or as small as possible.	$180x_S + 80x_G$, minimised
Constraints	Equations and inequalities every admissible answer must satisfy.	$40x_S + 15x_G \geq 600$; $x_S \leq 12$; $x_G \leq 20$; $x_S, x_G \geq 0$
Assumptions	Simplifications that make the problem solvable, and that you should be prepared to defend.	every shift completes its full quota; costs do not change with volume; shifts can be fractional

That last assumption deserves a comment, because it is the kind of thing a model quietly does to you. Nothing in the mathematics stops x_G from coming out as 8.4, and there is no such thing as 0.4 of a shift. Here we were lucky and the answer landed on whole numbers. In general, forcing variables to be whole numbers makes a problem substantially harder to solve — it becomes an *integer* program. Recognising when you have made this assumption is more important right now than knowing how to fix it.

Common pitfall

Confusing the objective with a constraint. “Cover 600 deliveries” and “spend as little as possible” are doing different jobs. The first is a bar you must clear — a constraint, written with $\geq$. The second is what you optimise — the objective, with no number on the right-hand side.

A quick diagnostic: a constraint has a comparison symbol ($\leq, \geq, =$) and a fixed number. An

objective has the word *minimise* or *maximise* and no comparison at all. If you have written “minimise cost ≤ 2800 ”, you have merged the two.

2.3 Deterministic and probabilistic models

The courier model is **deterministic**: every parameter is treated as known with certainty. Put the same inputs in twice and you get the same answer twice.

That is a strong assumption, and often a false one. The forecast of 600 deliveries is a *forecast*; the true number will be something else. A **probabilistic** (or *stochastic*) model replaces a fixed value with a random quantity:

$$D \sim \mathcal{N}(600, 50^2),$$

read as “ D follows a normal distribution with mean 600 and standard deviation 50”.

⚠ Common pitfall

In the notation $\mathcal{N}(\mu, \sigma^2)$, the **second argument is the variance, not the standard deviation**. Writing $\mathcal{N}(600, 50^2)$ means a standard deviation of 50. Writing $\mathcal{N}(600, 50)$ means a *variance* of 50, hence a standard deviation of about 7.07 — a completely different model. Some software (including R’s `rnorm`) takes the standard deviation as its argument instead, which is a frequent source of error when moving between a formula and code.

Once demand is random, the question changes shape. “Cover all demand” is no longer something you can guarantee; you can only make it likely. Questions become “what schedule covers demand at least 95% of the time?” That is beyond this week, but it is worth seeing where the road leads.

2.4 Statistical models

A third kind of model does not choose anything. It explains: given data, how does one quantity move with another?

💡 Worked example: how long does a delivery take?

A logistics team wants to predict delivery time from distance. For delivery i they record the time T_i in minutes and the distance d_i in kilometres, and propose

$$T_i = \beta_0 + \beta_1 d_i + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2).$$

Reading the pieces:

- β_0 is the time a delivery takes even at zero distance — parking, finding the door, hand-over.
- β_1 is the additional minutes per extra kilometre.

- ε_i is everything else: traffic that day, a slow lift, a customer who does not answer. It is not an error in the sense of a mistake; it is the part of T_i that distance does not explain.
- σ^2 measures how much that unexplained part varies.

This is where the vocabulary of *dependent* and *independent* variables belongs, and it is worth defining properly rather than mentioning in passing.

i Definition: response and explanatory variables

In a statistical model, the **response variable** (also called the **dependent variable**) is the quantity the model sets out to explain or predict. It sits on the left-hand side.

An **explanatory variable** (also called an **independent variable**, a *predictor*, a *regressor*, or a *covariate*) is a quantity used to account for variation in the response. Explanatory variables sit on the right-hand side.

In words: the response is what you want to know; the explanatory variables are what you already know and are using to guess it.

In the delivery model, T_i is the response and d_i is the explanatory variable. The names “dependent” and “independent” are older and, frankly, misleading — “independent” here has nothing to do with statistical independence, and the explanatory variables are frequently related to one another. This course prefers *response* and *explanatory*, but you will meet all of these words in the wild and they mean the same thing.

! Common pitfall

Do not use “response” and “explanatory” for a model that has no random component. They are statistical terms; they presuppose data and unexplained variation. In a deterministic model such as a supply-and-demand equilibrium, where you solve algebraically for a price, the right distinction is between quantities determined *inside* the model (**endogenous**) and quantities taken as given from *outside* it (**exogenous**). Calling the price an “explanatory variable” when the whole point is to solve for it is a contradiction — it cannot both explain and be the unknown.

Note also the notation split established in the conventions table, which now has a reason attached: $\beta_0, \beta_1, \sigma^2$ are properties of the *population* and are unknown; the numbers you compute from a sample are estimates of them, written $\hat{\beta}_0, \hat{\beta}_1, \hat{\sigma}^2$. The same distinction separates μ (a population mean) from $\bar{x}$ (a sample mean). They are different objects and the notation should never blur them.

3 Sets

A **set** is a collection of distinct objects, called its **elements**. That is the whole definition. Sets are useful in analytics because so many things are naturally collections: the customers in a segment, the days in a quarter, the values a variable is permitted to take, the decisions that satisfy every constraint.

We write $x \in A$ for “ x is an element of A ”, and $x \notin A$ for “ x is not”.

3.1 Three ways to describe a set

i Definition: the three descriptions

By listing (roster notation): write the elements between braces.

$$A = \{2, 4, 6, 8\}$$

By a defining property (set-builder notation): state the universe the elements are drawn from, a vertical bar read “such that”, and the condition.

$$B = \{x \in \mathbb{Z} \mid x \text{ is even and } 2 \leq x \leq 8\}$$

By an interval: for sets of consecutive real numbers only.

$$C = [2, 8]$$

In words: you can name the members one by one, state the rule for membership, or — if the members form an unbroken stretch of the number line — give the two endpoints.

Note that A and B above are the *same set*, written two ways. C is a different set: it contains 3.7, which A does not.

Listing is only practical for small finite sets. Set-builder notation is the workhorse, and it is worth being fussy about its three parts, because omitting the universe is the most common error in the whole topic:

$$\underbrace{\{x \in \mathbb{R} \mid}_{\text{where } x \text{ lives}} \underbrace{x \geq 0\}_{\text{the condition}} \}$$

Write $\{x \mid x \geq 0\}$ without the $\in \mathbb{R}$ and you have not said whether you mean the non-negative integers, the non-negative reals, or something else. The universe is not decoration.

3.2 The standard number sets

Notation	Name	Contains	Typical use
$\mathbb{N}$	naturals	$0, 1, 2, 3, \dots$	counts: customers, orders, periods
$\mathbb{Z}$	integers	$\dots, -2, -1, 0, 1, 2, \dots$	changes that can go either way
$\mathbb{Q}$	rationals	all p/q with $p, q \in \mathbb{Z}$, $q \neq 0$	exact proportions
$\mathbb{R}$	reals	everything on the number line	prices, times, rates
$\mathbb{R}_+$	non-negative reals	$[0, +\infty)$	quantities that cannot be negative
$\mathbb{R}^p$	real p -vectors	ordered lists of p reals	one observation, p variables
$\mathbb{R}^{n \times p}$	real matrices	n rows, p columns	a whole dataset

These nest: $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$. Every integer is a rational (write -3 as $-3/1$), and every rational is real. The reverse fails: $\sqrt{2}$ and π are real but not rational.

Common pitfall

$0 \in \mathbb{N}$ in this course. Many textbooks start the naturals at 1. Ours starts at 0, so the statement “0 is a natural number” is true here. This is a convention, not a fact about the world, and conventions differ — which is exactly why the course fixes one and states it.

Relatedly, $\mathbb{R}_+ = [0, +\infty)$ **includes** zero. If you need strictly positive reals, write $(0, +\infty)$ and say so.

3.3 Intervals

An interval is the set of all reals between two endpoints. The only thing to get right is whether the endpoints themselves are in.

Notation	Set-builder form	Endpoints included?
$[a, b]$	$\{x \in \mathbb{R} \mid a \leq x \leq b\}$	both
(a, b)	$\{x \in \mathbb{R} \mid a < x < b\}$	neither
$[a, b)$	$\{x \in \mathbb{R} \mid a \leq x < b\}$	left only
$(a, b]$	$\{x \in \mathbb{R} \mid a < x \leq b\}$	right only
$[a, +\infty)$	$\{x \in \mathbb{R} \mid x \geq a\}$	left only
$(-\infty, b]$	$\{x \in \mathbb{R} \mid x \leq b\}$	right only

The pattern: **a square bracket admits the endpoint, a round bracket excludes it**. Infinity always gets a round bracket, because $+\infty$ is not a number and cannot be an element of anything.

 Common pitfall

You may see the open interval written $]a, b[$, with the brackets turned outwards. This is the standard French convention and means exactly the same thing as (a, b) . Some of the course slides use it. This course, and essentially all statistical software, uses (a, b) .

3.4 Absolute value and distance

 Definition: absolute value

For $a \in \mathbb{R}$,

$$|a| = \begin{cases} a, & \text{if } a \geq 0, \\ -a, & \text{if } a < 0. \end{cases}$$

In words: strip the sign. The absolute value of a number is how far it is from zero, and distance is never negative.

So $|7| = 7$, $|-7| = 7$, and $|0| = 0$. Note that $-a$ is *positive* when a is negative: $|-7| = -(-7) = 7$. The minus sign in the definition is not a typo.

 Key result: distance and tolerance

The distance between two reals u and v is

$$d(u, v) = |u - v|,$$

and for any $c > 0$,

$$|u - v| \leq c \iff -c \leq u - v \leq c \iff v - c \leq u \leq v + c \iff u \in [v - c, v + c].$$

In words: “ u is within c of v ” and “ u lies in the interval of width $2c$ centred at v ” say the same thing. Absolute-value inequalities are interval statements in disguise.

This one identity converts a large class of business requirements into intervals you can compute with.

 Worked example: a cold-chain shipment

A vaccine shipment must be held at 5°C , and the carrier guarantees the temperature will never stray more than 2°C from target. Write the admissible temperature range as an interval.

Let T be the temperature. “Never more than 2 degrees from 5” is a statement about distance:

$$|T - 5| \leq 2.$$

Unfold it using the result above:

$$-2 \leq T - 5 \leq 2.$$

Add 5 throughout — adding the same number to all three parts of a chained inequality is always legal and does not flip anything:

$$\begin{aligned} -2 + 5 &\leq T \leq 2 + 5, \\ 3 &\leq T \leq 7. \end{aligned}$$

So $T \in [3, 7]$. The endpoints are included because the guarantee was “no more than”, written $\leq$; had it been “strictly less than”, we would have written $(3, 7)$.

3.5 Subsets and cardinality

i Definition: subset

A is a **subset** of B , written $A \subseteq B$, if every element of A is also an element of B :

$$A \subseteq B \iff (\text{for every } x, x \in A \Rightarrow x \in B).$$

In words: A sits entirely inside B . Nothing in A escapes.

A natural example: among the recipients of an email campaign, let O be those who opened it and K those who clicked a link inside it. You cannot click a link in an email you never opened, so $K \subseteq O$. Note this is a claim about *every possible dataset*, not about one particular week’s numbers — that is what makes it a subset relation rather than a coincidence.

i Definition: cardinality

For a finite set A , the **cardinality** $|A|$ is the number of elements in A .

So $|\{2, 4, 6, 8\}| = 4$ and $|\emptyset| = 0$, where $\emptyset$ denotes the **empty set**, the set with no elements at all.

3.6 Set operations

Let A and B be subsets of a **universe** Ω — the reference set of everything currently under consideration.

Operation	Notation	Definition	Reading
Union	$A \cup B$	$\{x \in \Omega \mid x \in A \text{ or } x \in B\}$	in at least one
Intersection	$A \cap B$	$\{x \in \Omega \mid x \in A \text{ and } x \in B\}$	in both
Difference	$A \setminus B$	$\{x \in \Omega \mid x \in A \text{ and } x \notin B\}$	in A but not B
Complement	A^c	$\{x \in \Omega \mid x \notin A\}$	everything outside A

The “or” in the union is **inclusive**: x may be in A , in B , or in both. Everyday English often uses “or” exclusively (“tea or coffee” usually rules out both), which is a genuine source of confusion. Mathematical “or” never does.

Two relations worth noticing: $A \setminus B = A \cap B^c$, and $A^c = \Omega \setminus A$. Difference and complement are the same operation, one relative to B and one relative to the whole universe.

 Worked example: two campaign audiences

Ten subscribers, numbered 1 to 10, received a product announcement. Let

$$\Omega = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

be all recipients, and record two behaviours:

$$O = \{1, 3, 4, 6, 9, 10\} \quad (\text{opened the email}),$$

$$P = \{2, 3, 4, 7, 9\} \quad (\text{visited the pricing page}).$$

Compute each operation by reading its definition literally.

Intersection — in both lists: 3, 4, 9 appear in each.

$$O \cap P = \{3, 4, 9\}.$$

Union — in at least one list; collect everything, listing each subscriber once.

$$O \cup P = \{1, 2, 3, 4, 6, 7, 9, 10\}.$$

Difference $O \setminus P$ — opened but did not reach pricing. From O , remove 3, 4, 9.

$$O \setminus P = \{1, 6, 10\}.$$

Difference $P \setminus O$ — reached pricing without opening the email. From P , remove 3, 4, 9.

$$P \setminus O = \{2, 7\}.$$

Complement — everyone who did not open.

$$O^c = \{2, 5, 7, 8\}.$$

Interpreting them. $O \cap P$ is the engaged group: the campaign plausibly drove them to pricing. $P \setminus O$ is more interesting than it looks — two people reached the pricing page without opening the email, so they arrived by some other route. Attributing their visit to the campaign would overstate its effect. Notice that O and P overlap without either containing the other, so neither $O \subseteq P$ nor $P \subseteq O$ holds.

 Key result: the size of a union (inclusion–exclusion)

For finite sets A and B ,

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

In words: add the two sizes, then subtract the overlap once — because anything in both sets

got counted twice when you added.

Checking it on the campaign data: $|O| = 6$, $|P| = 5$, $|O \cap P| = 3$, so the formula predicts $6 + 5 - 3 = 8$. Counting $O \cup P = \{1, 2, 3, 4, 6, 7, 9, 10\}$ directly gives 8, which matches.

 Common pitfall

Adding audience sizes without subtracting the overlap. “The email reached 6 people and the pricing page had 5 visitors, so 11 people engaged” is wrong: it is 8. This error is endemic in marketing reports, where it inflates reach, and it is exactly what inclusion–exclusion corrects. The two counts only add cleanly when $A \cap B = \emptyset$.

3.7 Cartesian products and the data matrix

 Definition: Cartesian product

The **Cartesian product** of A and B is the set of ordered pairs with first component from A and second from B :

$$A \times B = \{(a, b) \mid a \in A, b \in B\}.$$

In words: every possible pairing of something from A with something from B . Order matters — (a, b) and (b, a) are different pairs.

Repeating this gives $\mathbb{R}^p$: an element of $\mathbb{R}^p$ is an ordered list of p real numbers. This is precisely how a single observation with p measured variables is represented.

Stack n such observations and you have a dataset:

$$X = (x_{ij}) \in \mathbb{R}^{n \times p},$$

where x_{ij} is the value of variable j for observation i . Row i is one observation; column j is one variable across all observations.

 Worked example: reading a data matrix

Four subscribers are described by three variables — tenure in months, monthly spend in dollars, and support tickets raised:

$$X = \begin{pmatrix} 8 & 42 & 1 \\ 24 & 115 & 0 \\ 3 & 28 & 4 \\ 36 & 96 & 2 \end{pmatrix}.$$

Which space does X belong to? Four rows, three columns, so $X \in \mathbb{R}^{4 \times 3}$. Here $n = 4$ observations and $p = 3$ variables.

What is x_{32} ? Row 3, column 2. Row 3 is $(3, 28, 4)$, and its second entry is 28 — subscriber 3 spends \$28 per month. The row index always comes first.

What is observation 2? The second row, (24, 115, 0): a subscriber of 24 months, spending \$115, who has never contacted support.

What is the mean of variable 2? Add the second column and divide by the number of observations:

$$\frac{42 + 115 + 28 + 96}{4} = \frac{281}{4} = 70.25.$$

Average monthly spend is \$70.25.

⚠ Common pitfall

Reading n and p off the wrong axis. A dataset in $\mathbb{R}^{100 \times 8}$ has 100 observations and 8 variables, not the other way round. Rows are observations; columns are variables. In these notes n *always* means the number of observations and p *always* means the number of variables, and no other letter is used for either.

3.8 Partitions

i Definition: partition

A family of sets $X_1, X_2, \dots, X_k$ is a **partition** of the universe Ω if all three of the following hold:

1. each X_i is non-empty: $X_i \neq \emptyset$;
2. they do not overlap: $X_i \cap X_j = \emptyset$ whenever $i \neq j$;
3. together they account for everything: $\bigcup_{i=1}^k X_i = \Omega$.

In words: the sets carve Ω into pieces with no gaps and no overlaps — every element of Ω lands in exactly one piece.

The phrase “exactly one” is the whole content of the definition. Condition 3 gives *at least one*; condition 2 gives *at most one*. Together: exactly one.

Partitions are what a categorical variable does to a dataset. Assigning every customer a tier, every transaction a region, or every ticket a priority partitions the population — provided the categories are built correctly.

💡 Worked example: customer tiers

A retailer sorts customers by annual spend s into Bronze, Silver, and Gold:

$$T_1 = \{c \in \Omega \mid s(c) < 500\}, \quad T_2 = \{c \in \Omega \mid 500 \leq s(c) < 2000\}, \quad T_3 = \{c \in \Omega \mid s(c) \geq 2000\}.$$

Is this a partition? Check the conditions.

No overlaps. Take a customer with $s(c) = 500$. Is it in T_1 ? That needs $500 < 500$, which is

false. In T_2 ? That needs $500 \leq 500 < 2000$, true. In T_3 ? Needs $500 \geq 2000$, false. Exactly one. The same check at the other boundary, $s(c) = 2000$, puts the customer in T_3 only.

Covers everything. Any spend s is either below 500, or at least 500 but below 2000, or at least 2000 — no fourth possibility.

Non-empty. Provided the retailer has at least one customer in each band.

So $\{T_1, T_2, T_3\}$ partitions the customer base.

⚠ Common pitfall

Boundary values assigned to two tiers. Had the retailer written the bands as $[0, 500]$, $[500, 2000]$, $[2000, \infty)$ — square brackets throughout — then a customer spending exactly \$500 would qualify for both Bronze and Silver, and one spending \$2000 for both Silver and Gold. That is not a partition, and in practice it means the tiers sum to more than the customer base and a customer's tier depends on which rule the query happens to evaluate first.

The fix is the pattern used above: **closed on the left, open on the right**, $[a, b)$, for every band except the last. This is the convention nearly all statistical software follows when binning a continuous variable, and it is worth adopting as a habit.

3.9 The simplex

One particular set recurs so often — in weights, proportions, market shares, and predicted class probabilities — that it has a name.

i Definition: the simplex

The **simplex** in $\mathbb{R}^p$ is

$$\Delta_p = \left\{ \mathbf{w} \in \mathbb{R}_+^p \mid \sum_{j=1}^p w_j = 1 \right\}.$$

In words: the set of vectors with p components that are all non-negative and add up to exactly 1 — that is, the set of all possible ways of splitting a whole into p shares.

Two conditions, and both are needed. Non-negativity alone permits shares summing to 3; summing to 1 alone permits a negative share. The subscript p counts *components*.

💡 Worked example: market shares

Three firms hold shares $\mathbf{v} = (0.45, 0.35, 0.20)$. Is $\mathbf{v} \in \Delta_3$?

Non-negative? $0.45 \geq 0$, $0.35 \geq 0$, $0.20 \geq 0$. Yes.

Sums to one? $0.45 + 0.35 + 0.20 = 1.00$. Yes.

Both conditions hold, so $\mathbf{v} \in \Delta_3$.

Now test $\mathbf{u} = (0.60, 0.55, -0.15)$. Its components sum to $0.60 + 0.55 - 0.15 = 1.00$, so the second condition passes. But $-0.15 < 0$, so the first fails, and $\mathbf{u} \notin \Delta_3$. A negative market share is not a rounding quibble — it is meaningless, and the non-negativity condition is what rules it out.

 Common pitfall

Superscript versus subscript on Δ . You will see Δ^2 , Δ_3 , and Δ^{K-1} in different sources, all describing three shares summing to one. The disagreement is over whether the index counts components (3) or dimensions (2 — a triangle is a two-dimensional object even though it sits in three-dimensional space). This course always uses the subscript and always counts **components**: Δ_p has p of them. When you read another source, check which it means before comparing.

4 Summation notation

Summation notation is not a new idea; it is shorthand for addition. It earns its place because it lets you write a sum whose length is a symbol rather than a number, and because it makes the algebra of sums visible.

4.1 The notation

i Definition: the summation symbol

$$\sum_{i=1}^n a_i = a_1 + a_2 + \cdots + a_n.$$

In words: “add up the a_i , letting i run from 1 to n .” The letter i is the **index**, the numbers below and above $\sum$ are the **limits**, and a_i is the **summand**.

The index is a placeholder with no meaning outside the sum. Writing $\sum_{j=1}^n a_j$ or $\sum_{k=1}^n a_k$ denotes exactly the same number. This matters more than it seems: an index that also appears *outside* the sum is a symptom of an error.

Reading in both directions is the skill to build:

$$\sum_{i=1}^4 x_i = x_1 + x_2 + x_3 + x_4, \quad \sum_{t=0}^3 r_t = r_0 + r_1 + r_2 + r_3,$$
$$\sum_{j=2}^5 (2j-1) = 3 + 5 + 7 + 9 = 24, \quad \sum_{k=1}^n c = \underbrace{c + c + \cdots + c}_{n \text{ terms}} = nc.$$

That last one is worth staring at. The summand c does not mention k , so nothing changes as k advances; you simply add c to itself n times. Students routinely write $\sum_{k=1}^n c = c$, forgetting that the sum still runs.

4.2 Three rules

! Key result: algebra of sums

For any numbers a_i, b_i and any constant c (a value not depending on the index):

$$\textbf{Additivity: } \sum_{i=1}^n (a_i + b_i) = \sum_{i=1}^n a_i + \sum_{i=1}^n b_i$$

$$\textbf{Homogeneity: } \sum_{i=1}^n c a_i = c \sum_{i=1}^n a_i$$

$$\textbf{Constant: } \sum_{i=1}^n c = nc$$

In words: sums split across addition, constants slide out front, and a constant summand just gets multiplied by the number of terms.

The first two together are called **linearity**, and they are the reason summation notation is worth learning: they let you manipulate sums without ever expanding them.

! Common pitfall

There is no product rule and no rule for squares. In general

$$\sum a_i b_i \neq \left(\sum a_i \right) \left(\sum b_i \right), \quad \sum a_i^2 \neq \left(\sum a_i \right)^2.$$

A two-term counterexample settles it: with $a = (1, 2)$, $\sum a_i^2 = 1 + 4 = 5$ while $(\sum a_i)^2 = 3^2 = 9$. Linearity covers addition and constant multiples — nothing else.

4.3 The mean and deviations from it

i Definition: arithmetic mean

For observed values $x_1, \dots, x_n$, the **sample mean** is

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i.$$

The quantity $x_i - \bar{x}$ is the **deviation** of observation i from the mean.

Multiplying both sides by n gives a form used constantly in derivations:

$$\sum_{i=1}^n x_i = n\bar{x}.$$

! Key result: deviations sum to zero

$$\sum_{i=1}^n (x_i - \bar{x}) = 0.$$

Derivation. Split the sum using additivity, then handle each piece:

$$\sum_{i=1}^n (x_i - \bar{x}) = \sum_{i=1}^n x_i - \sum_{i=1}^n \bar{x}.$$

The second sum has a constant summand — $\bar{x}$ is a single fixed number, not something that changes with i — so by the constant rule it equals $n\bar{x}$:

$$= \sum_{i=1}^n x_i - n\bar{x}.$$

But $\sum x_i = n\bar{x}$, so this is

$$= n\bar{x} - n\bar{x} = 0. \quad \blacksquare$$

In words: the mean sits exactly at the balance point of the data. Whatever lies above it is matched, in total, by what lies below. This is why raw deviations are useless as a measure of spread — they always total zero, no matter how scattered the data — and hence why we square them.

! Key result: the computational form of the sum of squares

$$\sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n x_i^2 - n\bar{x}^2.$$

Derivation. Expand the square inside the sum:

$$\sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n (x_i^2 - 2x_i\bar{x} + \bar{x}^2).$$

Split into three sums by additivity:

$$= \sum_{i=1}^n x_i^2 - \sum_{i=1}^n 2x_i\bar{x} + \sum_{i=1}^n \bar{x}^2.$$

In the middle sum, $2\bar{x}$ is constant, so pull it out by homogeneity. In the last, $\bar{x}^2$ is constant, so apply the constant rule:

$$= \sum_{i=1}^n x_i^2 - 2\bar{x} \sum_{i=1}^n x_i + n\bar{x}^2.$$

Substitute $\sum x_i = n\bar{x}$ into the middle term:

$$= \sum_{i=1}^n x_i^2 - 2\bar{x}(n\bar{x}) + n\bar{x}^2 = \sum_{i=1}^n x_i^2 - 2n\bar{x}^2 + n\bar{x}^2.$$

Combine the last two terms:

$$= \sum_{i=1}^n x_i^2 - n\bar{x}^2. \quad \blacksquare$$

In words: you can get the sum of squared deviations without ever computing a single deviation — square the raw values, add them, and subtract n times the squared mean. Before computers this saved enormous labour. It is still how the formula is often stored in software, though it is numerically less stable than the direct form when the data are large and close together.

Worked example: weekly order volumes

A warehouse records orders over five weeks:

$$\mathbf{x} = (120, 95, 140, 105, 90).$$

The mean.

$$\bar{x} = \frac{120 + 95 + 140 + 105 + 90}{5} = \frac{550}{5} = 110.$$

The deviations. Subtract 110 from each:

$$(10, -15, 30, -5, -20).$$

Their total: $10 - 15 + 30 - 5 - 20 = 0$, as the first key result guarantees. This is a useful arithmetic check — if your deviations do not sum to zero, the mean is wrong.

Sum of squared deviations, directly.

$$\sum (x_i - \bar{x})^2 = 10^2 + (-15)^2 + 30^2 + (-5)^2 + (-20)^2 = 100 + 225 + 900 + 25 + 400 = 1650.$$

The same quantity, computationally. First the raw squares:

$$\sum x_i^2 = 120^2 + 95^2 + 140^2 + 105^2 + 90^2 = 14400 + 9025 + 19600 + 11025 + 8100 = 62150.$$

Then

$$\sum x_i^2 - n\bar{x}^2 = 62150 - 5(110)^2 = 62150 - 5(12100) = 62150 - 60500 = 1650.$$

The two routes agree, as they must.

4.4 Double sums

When data are arranged in a grid, two indices are needed: a_{ij} for row i , column j .

$$\sum_{i=1}^m \sum_{j=1}^n a_{ij}$$

means: for each row i , add across the columns to get a row total; then add the row totals. The inner sum runs first.

! Key result: order of summation

For a finite grid,

$$\sum_{i=1}^m \sum_{j=1}^n a_{ij} = \sum_{j=1}^n \sum_{i=1}^m a_{ij}.$$

In words: totalling by rows and totalling by columns give the same grand total — you are adding the same numbers in a different order.

💡 Worked example: totalling a grid two ways

$$A = \begin{pmatrix} 3 & 1 & 4 \\ 2 & 5 & 0 \end{pmatrix}.$$

By rows.

$$S_1 = 3 + 1 + 4 = 8, \quad S_2 = 2 + 5 + 0 = 7, \quad S_1 + S_2 = 15.$$

By columns.

$$T_1 = 3 + 2 = 5, \quad T_2 = 1 + 5 = 6, \quad T_3 = 4 + 0 = 4, \quad T_1 + T_2 + T_3 = 15.$$

Both give 15. Nothing deep is happening — six numbers were added in two different orders — but the fact licenses swapping summation order in algebra, which is used constantly.

⚠️ Common pitfall

Reusing an index. In $\sum_i \sum_i a_{ii}$ the two sums collide and the expression is meaningless. Each sum needs its own index letter. Likewise, if i appears outside the sum in your final answer — as in “ $\sum_{i=1}^n x_i = nx_i$ ” — something has gone wrong: the index cannot survive the summation.

5 Algebra for working with models

This section is a toolkit. The operations are ones you have seen; what is worth rebuilding is the *fluency* to apply them without hesitation, because in later weeks they will be steps inside a larger argument rather than the point of the exercise.

5.1 Rearranging expressions

Operation	Identity	Use
Collect like terms	$2x + 3x = 5x$	tidy up
Expand	$a(b + c) = ab + ac$	remove brackets
Factor	$ab + ac = a(b + c)$	expose a common factor
Square of a sum	$(a + b)^2 = a^2 + 2ab + b^2$	expand quadratics
Difference of squares	$a^2 - b^2 = (a - b)(a + b)$	factor, simplify fractions

Common pitfall

$(a + b)^2 \neq a^2 + b^2$. The cross term $2ab$ is not optional. Test with $a = b = 1$: the left side is 4, and $a^2 + b^2 = 2$. The same error in a different costume: $\sqrt{a + b} \neq \sqrt{a} + \sqrt{b}$, and $\frac{1}{a + b} \neq \frac{1}{a} + \frac{1}{b}$.

5.2 Linear equations

A **linear equation in one unknown** has the form $ax + b = 0$ with $a \neq 0$, and the single solution

$$x = -\frac{b}{a}.$$

The condition $a \neq 0$ matters: if $a = 0$ the equation reads $b = 0$, which is either always true (if $b = 0$, so every x works) or never true (if $b \neq 0$, so no x works). Dividing by a without checking it is non-zero is the most common way to lose a case.

5.3 Quadratic equations

i Definition: quadratic equation and discriminant

A **quadratic equation** is $ax^2 + bx + c = 0$ with $a \neq 0$. Its **discriminant** is

$$\Delta = b^2 - 4ac.$$

⚠ Notation clash — read once

The letter Δ is doing two unrelated jobs in this course. Bare Δ , as here, is the **discriminant** of a quadratic — a single number. Subscripted Δ_p , from Section 2, is the **simplex** — a set of vectors. They have nothing to do with each other; the alphabet is simply short. The subscript is what tells them apart, so read it.

! Key result: solutions of a quadratic

- If $\Delta > 0$: two distinct real roots, $x = \frac{-b \pm \sqrt{\Delta}}{2a}$.
- If $\Delta = 0$: one repeated real root, $x = -\frac{b}{2a}$.
- If $\Delta < 0$: no real roots.

In words: the discriminant is a preview. Compute it first and you know how many answers to expect before doing any work — and whether a real answer exists at all.

The vertex of the parabola $y = ax^2 + bx + c$ sits at $x = -b/(2a)$, midway between the roots. When $a < 0$ the parabola opens downward and the vertex is a maximum — which is why quadratics show up in profit problems.

💡 Worked example: break-even and best output

A meal-kit company sells boxes at \$30 each. Producing q boxes per day costs

$$C(q) = 800 + 20q + 0.02q^2,$$

where \$800 is fixed daily cost, \$20 is the direct cost per box, and the quadratic term captures overtime and spoilage as the kitchen is pushed harder.

Build the profit function. Revenue is $R(q) = 30q$, so

$$\pi(q) = R(q) - C(q) = 30q - (800 + 20q + 0.02q^2).$$

Distribute the minus sign across every term in the bracket — this is where sign errors happen:

$$\pi(q) = 30q - 800 - 20q - 0.02q^2 = -0.02q^2 + 10q - 800.$$

Find the break-even points. Break-even means $\pi(q) = 0$:

$$-0.02q^2 + 10q - 800 = 0.$$

Here $a = -0.02$, $b = 10$, $c = -800$. The discriminant:

$$\Delta = b^2 - 4ac = 10^2 - 4(-0.02)(-800) = 100 - 64 = 36.$$

Note both minus signs in the product $4ac$: $(-0.02)(-800) = +16$, and $4 \times 16 = 64$, which is then *subtracted*. Since $\Delta = 36 > 0$ there are two roots, and $\sqrt{36} = 6$:

$$q = \frac{-10 \pm 6}{2(-0.02)} = \frac{-10 \pm 6}{-0.04}.$$

Taking the two signs separately:

$$q_1 = \frac{-10 + 6}{-0.04} = \frac{-4}{-0.04} = 100, \quad q_2 = \frac{-10 - 6}{-0.04} = \frac{-16}{-0.04} = 400.$$

Find the output that maximises profit. Since $a = -0.02 < 0$ the parabola opens downward, so the vertex is the maximum:

$$q^* = -\frac{b}{2a} = -\frac{10}{2(-0.02)} = -\frac{10}{-0.04} = 250,$$

with profit

$$\pi(250) = -0.02(250)^2 + 10(250) - 800 = -0.02(62500) + 2500 - 800 = -1250 + 1700 = 450.$$

Reading the answer. The company breaks even at 100 boxes and again at 400. Between those it makes money, peaking at 250 boxes with \$450 profit a day. Below 100 the \$800 fixed cost is not covered; above 400 the rising marginal cost overwhelms the \$30 price. The existence of an *upper* break-even point is the interesting part — it is easy to assume more output is always better, and the quadratic cost term is precisely what makes that false.

5.4 Exponents

For $a, b > 0$ and compatible exponents:

$$a^m a^n = a^{m+n}, \quad \frac{a^m}{a^n} = a^{m-n}, \quad (a^m)^n = a^{mn},$$

$$(ab)^n = a^n b^n, \quad a^{-n} = \frac{1}{a^n}, \quad a^0 = 1, \quad a^{1/n} = \sqrt[n]{a}.$$

And for a fixed base $a > 0$ with $a \neq 1$, exponentials are one-to-one:

$$a^f = a^g \iff f = g.$$

 Common pitfall

$a^m a^n = a^{m+n}$, **not** a^{mn} . Exponents add when you multiply powers and multiply only when you raise a power to a power. Check with small numbers whenever unsure: $2^2 \cdot 2^3 = 4 \cdot 8 = 32 = 2^5$, and $5 = 2 + 3$.

5.5 Logarithms

 Definition: logarithm

For base $a > 0$ with $a \neq 1$,

$$\log_a(x) = y \iff a^y = x.$$

The **natural logarithm** is the case $a = e$, written $\ln$:

$$\ln(x) = y \iff e^y = x.$$

In words: a logarithm answers “what power do I raise the base to, in order to get x ?” It is the exponential run backwards.

The rules follow directly from the exponent rules, since logarithms *are* exponents:

$$\ln(xy) = \ln x + \ln y, \quad \ln\left(\frac{x}{y}\right) = \ln x - \ln y, \quad \ln(x^r) = r \ln x,$$

$$\ln(1) = 0, \quad \ln(e) = 1, \quad \ln(e^x) = x, \quad e^{\ln x} = x \quad (x > 0).$$

The third rule — that a logarithm turns a power into a multiplication — is the one that does the work in this course. It is how an unknown stuck in an exponent gets brought down where you can solve for it.

 Common pitfall

Two distinct traps.

The notation. “log” is ambiguous. In R and most mathematical writing, $\log(x)$ means the natural logarithm; in many spreadsheets, engineering texts, and calculators, **LOG** means base 10. These notes always write $\ln$ for the natural log and always show the base otherwise. When reading anything else, check.

The algebra. There is no rule for the logarithm of a sum:

$$\ln(x + y) \neq \ln x + \ln y.$$

The left side cannot be simplified at all. What $\ln x + \ln y$ equals is $\ln(xy)$ — a product, not a sum. This is probably the single most common algebra error in the course.

5.6 Growth and decay

! Key result: compound growth

A quantity starting at Q_0 and changing by $p\%$ per period has, after t periods,

$$Q(t) = Q_0 \left(1 + \frac{p}{100}\right)^t.$$

The quantity $1 + p/100$ is the **growth factor**. Growth uses $p > 0$; decay is the same formula with $p < 0$, giving a factor below 1.

The reason the factor is *multiplied* repeatedly rather than added is that each period's change applies to the new total, not the original — that is what “compound” means.

💡 Worked example: subscriber growth and doubling time

A service has 12,000 subscribers, growing 8% per year.

Subscribers after five years. The growth factor is $1 + 8/100 = 1.08$, so

$$Q(5) = 12000 \times (1.08)^5 = 12000 \times 1.46933 \approx 17,632.$$

When does the base double? We need $Q(t) = 24000$:

$$12000 \times (1.08)^t = 24000.$$

Divide both sides by 12,000 — note that the initial size cancels, so doubling time does not depend on where you start:

$$(1.08)^t = 2.$$

The unknown is in the exponent, so take natural logarithms of both sides:

$$\ln((1.08)^t) = \ln 2.$$

Apply $\ln(x^r) = r \ln x$ to bring t down:

$$t \ln(1.08) = \ln 2,$$

$$t = \frac{\ln 2}{\ln 1.08} = \frac{0.6931}{0.07696} \approx 9.01 \text{ years.}$$

So the subscriber base doubles in just over nine years.

💡 Worked example: depreciation

Equipment bought for \$50,000 loses 20% of its value each year. Its value after four years: Here $p = -20$, so the factor is $1 - 20/100 = 0.8$, and

$$V(4) = 50000 \times (0.8)^4 = 50000 \times 0.4096 = \$20,480.$$

A common wrong answer is $50000 \times (1 - 0.8) = \$10,000$, from subtracting 80% once instead of retaining 80% four times. Sanity-check by stepping through: $\$50,000 \rightarrow \$40,000 \rightarrow \$32,000 \rightarrow \$25,600 \rightarrow \$20,480$.

5.7 Turning a multiplicative model into a linear one

Many business relationships are multiplicative rather than additive, and logarithms are what make them tractable.

! Key result: log-linearisation

The multiplicative model

$$Y = AX^\beta e^{\gamma Z}, \quad A > 0, X > 0, Y > 0$$

becomes, on taking natural logarithms of both sides,

$$\ln Y = \ln A + \beta \ln X + \gamma Z.$$

In words: a model that multiplies its inputs turns into one that adds them, once you work in logarithms. The right-hand side is now a straight line in $\ln X$ and Z — which is exactly the form that standard regression can fit.

Derivation. Take $\ln$ of both sides and apply the product rule twice:

$$\ln Y = \ln(AX^\beta e^{\gamma Z}) = \ln A + \ln(X^\beta) + \ln(e^{\gamma Z}).$$

Now $\ln(X^\beta) = \beta \ln X$ by the power rule, and $\ln(e^{\gamma Z}) = \gamma Z$ since $\ln$ and e undo each other:

$$\ln Y = \ln A + \beta \ln X + \gamma Z. \quad \blacksquare$$

The parameter β has a clean interpretation here: it is the percentage change in Y associated with a one-percent change in X — an **elasticity**. That is why this functional form is ubiquitous in demand modelling.

💡 Worked example: isolating a variable

Given $Y = AX^\beta e^{-\gamma Z}$ with $A, X, Y > 0$ and $\beta \neq 0$, solve for X .

Start by isolating the power of X . Divide both sides by A and multiply by $e^{\gamma Z}$:

$$X^\beta = \frac{Y e^{\gamma Z}}{A}.$$

The right-hand side is strictly positive (a positive number divided by a positive number, times a positive exponential), so raising both sides to the power $1/\beta$ is legitimate:

$$X = \left(\frac{Y e^{\gamma Z}}{A} \right)^{1/\beta}.$$

Numerically, with $A = 12$, $\beta = 0.8$, $\gamma = 0.15$, $Z = 4$, $Y = 30$:

$$X = \left(\frac{30 e^{0.15 \times 4}}{12} \right)^{1/0.8} = \left(\frac{30 e^{0.6}}{12} \right)^{1.25}.$$

Since $e^{0.6} \approx 1.8221$:

$$X = \left(\frac{30 \times 1.8221}{12} \right)^{1.25} = \left(\frac{54.664}{12} \right)^{1.25} = (4.5553)^{1.25} \approx 6.655.$$

6 Proof techniques

i Covered in class

This section is developed during the lecture, not before it. It covers propositions and quantifiers ($\forall$, $\exists$, $\exists!$), implication and equivalence, necessary and sufficient conditions, and the five proof strategies: direct proof, contrapositive, contradiction, induction, and disproof by counterexample.

Notes will be added here after the session.

One thing is worth flagging in advance, because it trips up anyone who has written code. The symbol $\exists!$ means “**there exists exactly one**” — the $!$ marks *uniqueness*, not negation. It is a stronger claim than $\exists$, not its opposite:

$\exists x P(x)$ means at least one x works;

$\exists! x P(x)$ means exactly one x works.

For instance $\exists x \in \mathbb{R}$ with $x^2 = 4$ is true (both 2 and -2 qualify), but $\exists! x \in \mathbb{R}$ with $x^2 = 4$ is **false**, precisely because there are two. Negation is written $\neg$ or $\nexists$.

One-minute recap

🔥 If you remember nothing else

- **Variable or parameter?** Ask whether you know the number before you start solving. Known in advance $\Rightarrow$ parameter. The same symbol can switch roles when the question changes.
- **Set-builder notation needs a universe.** $\{x \in \mathbb{R} \mid x \geq 0\}$, not $\{x \mid x \geq 0\}$.
- **Brackets:** square includes the endpoint, round excludes it. Infinity always gets round.
- $|u-v| \leq c$ is the interval $[v-c, v+c]$. Every tolerance statement is an interval statement.
- $|A \cup B| = |A| + |B| - |A \cap B|$ — subtract the overlap, or you double-count.
- **A partition means exactly one.** Use $[a, b)$ bands so boundary values land in one bin only.
- $\sum$ is **linear and nothing more:** it splits over $+$ and lets constants out. There is no product rule.
- $\sum(x_i - \bar{x}) = 0$ always. Use it as an arithmetic check on any mean you compute.
- **Logs turn powers into products and products into sums** — that is how you get an unknown out of an exponent. But $\ln(x + y)$ does not simplify.
- **In this course:** $0 \in \mathbb{N}$; n = observations, p = variables; vectors are bold; $\ln$ is natural log.

Where to practise

Work the pre-class exercise set for Session 01 before reading its solutions. Exercises 1–5 cover the notation in Sections 2; Exercises 6–8 cover Section 3; Exercises 9–12 cover Sections 1 and 4. The tutorial sheet then extends into inequalities and set identities.

If you are short of time, the highest-value three are the summation identities (the two derivations in Section 3 are the ones that recur all semester), the partition boundary check, and the log-linearisation.